CO3 – AT2

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Case Study

Case Study 1: Customer Segmentation using K-Means and PCA

Problem Understanding
The retail store needs to divide customers into meaningful groups based on their **Annual Income** and **Spending Score**. This helps the company identify customers with similar purchasing behaviour and design targeted marketing campaigns.
Concept Applied
K-Means Clustering is an unsupervised learning technique that divides data into a specified number of clusters. Each customer is assigned to the cluster whose centroid is closest to the customer.
PCA (Principal Component Analysis) reduces the dimensionality of the data by transforming the original features into principal components. It is mainly used here to represent and visualize the customer clusters in two dimensions.
Analysis
The Annual Income and Spending Score values are first standardized so that both features have comparable influence. The **Elbow Method** is used to compare different numbers of clusters using WCSS. The point where the reduction in WCSS becomes smaller indicates a suitable number of clusters.
After selecting **3 clusters**, K-Means assigns customers according to their similarities. PCA then transforms the standardized data into two principal components, making the cluster distribution easier to visualize.
Solution Design
The combination of K-Means and PCA provides both **customer grouping and visual analysis**. The resulting clusters can represent different customer behaviours, such as customers with higher spending, lower spending, or moderate spending. These groups can be used for targeted advertisements and personalized offers.
Code

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

data = pd.DataFrame({
"Income":[15,15,16,16,17,17,18,18,19,19],
```

```
"Score":[39,81,6,77,40,76,6,94,3,72]
})

X = StandardScaler().fit_transform(data)

model = KMeans(n_clusters=3, random_state=42, n_init=10)
labels = model.fit_predict(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

print("Customer Segmentation")
print("Customers:", len(data))
print("Clusters: 3")

plt.scatter(X_pca[:,0], X_pca[:,1], c=labels)
plt.xlabel("PCA 1")
plt.ylabel("PCA 2")
plt.title("Customer Segmentation")
plt.show()
```

Output

```
Customer Segmentation
Customers: 10
Clusters: 3
```

Interpretation

The customers are successfully divided into three groups based on income and spending behaviour. PCA provides a two-dimensional representation of these groups. The segmentation can help the retail company develop different marketing strategies for each customer group.

Case Study 2: Dimensionality Reduction and GMM

Problem Understanding

The wine manufacturer has collected several chemical measurements such as **Alcohol, Malic Acid, Ash, Alcalinity, Magnesium, and Phenols**. Analyzing all these variables directly can make the data difficult to interpret. Dimensionality reduction is therefore applied before clustering.

Concept Applied

PCA reduces the number of dimensions while retaining maximum variance in the data.

Factor Analysis (FA) represents the observed variables using a smaller number of underlying factors.

ICA (Independent Component Analysis) transforms the data into statistically independent components.

Gaussian Mixture Model (GMM) is a probabilistic clustering method. It assumes that the data is generated from a mixture of Gaussian distributions. The **EM (Expectation-Maximization) algorithm** estimates the parameters of these distributions and assigns samples to clusters.

Analysis

The wine features are first standardized because they have different numerical scales. PCA, FA and ICA are then used to transform the six original features into two components.
GMM is applied to the PCA-transformed data. Unlike K-Means, GMM provides a probabilistic approach where each sample can have a probability of belonging to different clusters.

Solution Design

Dimensionality reduction makes the wine dataset easier to analyze while preserving important information. GMM can then identify groups of chemically similar wine samples. Comparing PCA, FA and ICA helps understand how different dimensionality-reduction methods represent the same data.

Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA, FactorAnalysis, FastICA
from sklearn.mixture import GaussianMixture

data = pd.DataFrame({
"Alcohol":[14.23,13.20,13.16,14.37,13.24,14.20,14.39,14.06,14.83,13.86],
"Malic":[1.71,1.78,2.36,1.95,2.59,1.76,1.87,2.15,1.64,1.35],
"Ash":[2.43,2.14,2.67,2.50,2.87,2.45,2.45,2.61,2.17,2.27],
"Alcalinity":[15.6,11.2,18.6,16.8,21,15.2,14.6,17.6,14,16],
"Magnesium":[127,100,101,113,118,112,96,121,97,98],
"Phenols":[2.80,2.65,2.80,3.85,2.80,3.27,2.50,2.60,2.80,2.98]
})

X = StandardScaler().fit_transform(data)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

fa = FactorAnalysis(n_components=2).fit_transform(X)
ica = FastICA(n_components=2, random_state=42).fit_transform(X)

gmm = GaussianMixture(n_components=3, random_state=42)
clusters = gmm.fit_predict(X_pca)

print("Wine Analysis")
print("Samples:", len(data))
print("Original features:", 6)
print("Reduced features:", 2)
print("GMM clusters:", 3)
```

Output

```
Wine Analysis
Samples: 10
Original features: 6
Reduced features: 2
GMM clusters: 3
```

Interpretation

The six chemical features are successfully reduced to two components using PCA, FA and ICA. GMM then groups the transformed wine samples into three clusters. The experiment demonstrates how dimensionality reduction simplifies high-dimensional data and how probabilistic clustering can be applied to the reduced representation.